

COMPUTER NETWORKS (24MC410)

I M.C.A II Semester

2025-26 EVEN SEMESTER

Name of the Faculty : Dr. K. Santhi Sri

Professor, Department of CA

VFSTR (Deemed to be University)

Institute Vision & Mission

VISION:

To evolve in to a centre of excellence in Science & Technology through creative and innovative practices in teaching-learning, towards promoting academic achievement and research excellence to produce internationally accepted, competitive and world class professionals who are psychologically strong & emotionally balanced imbued with social consciousness & ethical values.

MISSION:

To Provide High Quality academic programmes, training activities, research facilities and opportunities supported by continuous industry - institute interaction aimed at promoting employability, entrepreneurship, leadership and research aptitude among students and contribute to the economic and technological development of the region, state and nation.

Department Vision & Mission

VISION:

To emerge as a renowned center in the field of computer applications for providing high quality education and research to produce globally competent professionals with ethical values.

MISSION:

- **M1:** To provide high quality education with cutting-edge curriculum and learner-centered approach.
- **M2:** To establish the culture of innovation and research through industry- academia linkages.
- **M3:** To cultivate ethical and socially-conscious professionals with leadership qualities, who understand the global implications of technology.

PO's of the Department

- **PO1 (Foundation Knowledge):** Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.
- **PO2 (Problem Analysis):** Identify, review, formulate and analyze problems for primarily focusing on customer requirements using critical thinking frameworks.
- **PO3 (Development of Solutions):** Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.
- **PO4 (Modern Tool Usage):** Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO's of the Department

- **PO5 (Individual and Teamwork):** Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.
- **PO6 (Project Management and Finance):** Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.
- **PO7 (Ethics):** Commit to professional ethics in managing software projects with financial | aspects. Learn to use new technologies for cyber security and insulate customers from malware
- **PO8 (Life-long learning):** Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

PEO's of the Department

- **PEO1:** Develop cutting-edge technical skills to succeed in career and as an entrepreneur.
- **PEO2 :** Pursue higher education and research with critical thinking and problem solving capabilities
- **PEO3:** Demonstrate ethical, collaborative, and leadership abilities in multi-disciplinary domains.

PSO's of the Department

- PSO1: To provide solutions for complex computational problems with requisite mathematical, programming languages and tools.
- PSO2: To design, develop and analyse software and related services using algorithms, web design, and Big Data Analytics for real time applications.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on imparting knowledge about various protocols involved in LANs and WANs. In addition, it gives good foundation on different protocols such as data link protocols, internet protocols and transport protocols present in the respective layers of computer networks.

MODULE – 1

12L+12T+0P+12SL=36 Hours

UNIT – I

INTRODUCTION TO NETWORKS

Uses of Computer Networks, Types of Computer Networks, Network Technology From Local to Global, Examples of Networks, Network Protocols, Reference Models.

UNIT – II

PHYSICAL LAYER AND TRANSMISSION MEDIA

Guided Transmission Media: Twisted Pair, Coaxial, Fiber Optics, Wireless Transmission, Using the Spectrum for Transmission: Radio, Microwave, Infrared, Light transmission.

MODULE – 2

18L+18T+0P+18SL=54 Hours

UNIT – I

DATA LINK LAYER AND MAC LAYER

Data Link Layer Design Issues, Error Detection and Correction, Elementary Data Link Protocols, Sliding Window Protocols, The Channel Allocation Problem, Multiple Access Protocols, Ethernet, Wireless LANs.

UNIT – 2

THE NETWORK LAYER

Network Layer Design Issues, Routing Algorithms in A Single Network, Traffic Management At The Network Layer, Quality of Service and Application QOE, Internetworking, The Network Layer in the Internet (IPv4, IPv6, ICMP, ARP).

UNIT – 3

TRANSPORT AND APPLICATION LAYER

The Transport Service, Elements of Transport Protocols, Congestion Control, The Internet Transport Protocols: UDP, The Internet Transport Protocols: TCP **Application Layer** : DNS, Electronic Mail, The World Wide Web.

Syllabus

Module-1

UNIT-1

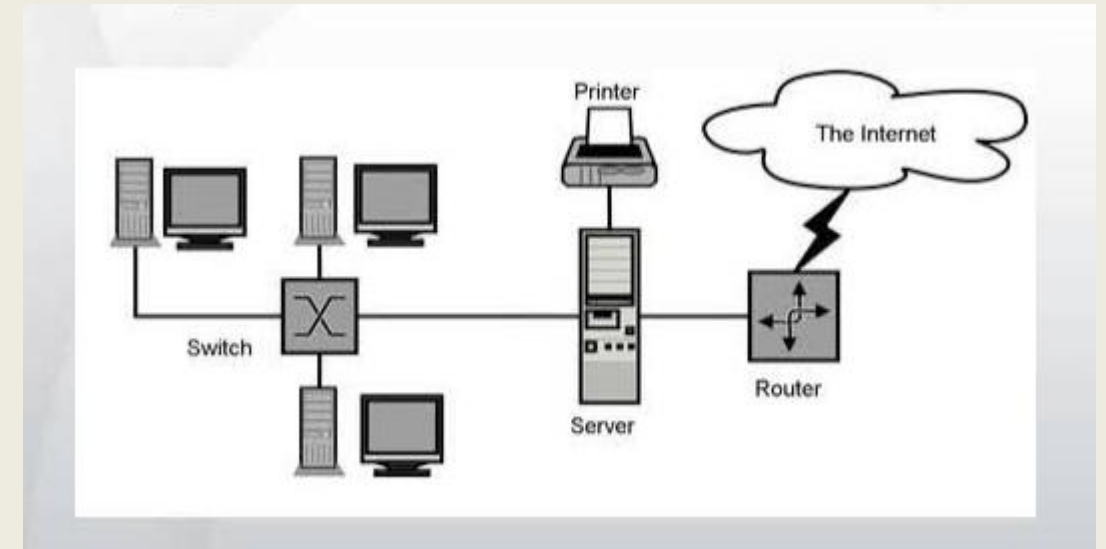
INTRODUCTION TO NETWORKS

Uses of Computer Networks, Types of Computer Networks, Network Technology From Local to Global, Examples of Networks, Network Protocols, Reference Models.

Uses of Computer Networks

Computer networks are used in many areas of modern life. Their major uses can be grouped into the following categories:

1. Access to Information
2. Person-to-Person Communication
3. Electronic Commerce (E-Commerce)
4. Entertainment
5. Internet of Things (IoT)



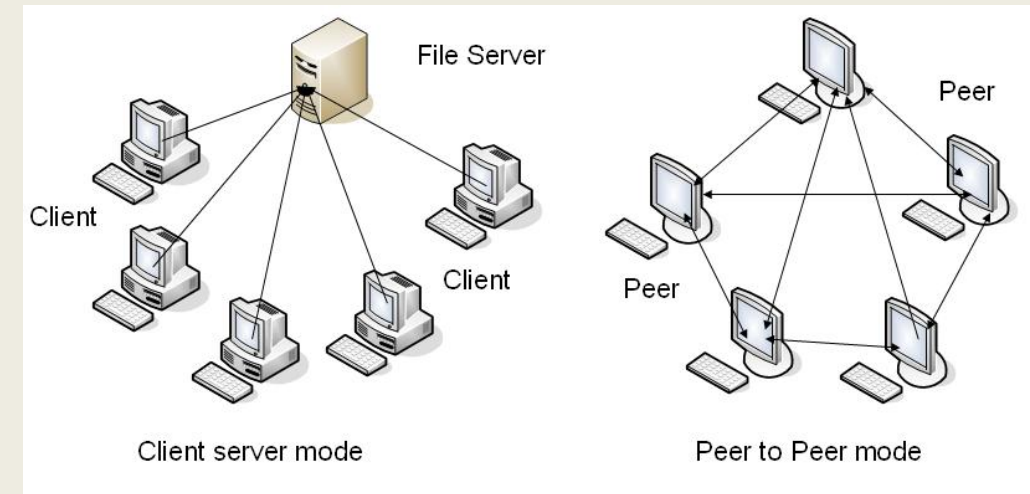
1. Access to Information

Computer networks make it possible for users to access information stored on remote systems. Instead of storing all data locally, information can be placed on servers and retrieved from anywhere through the network.

Examples:

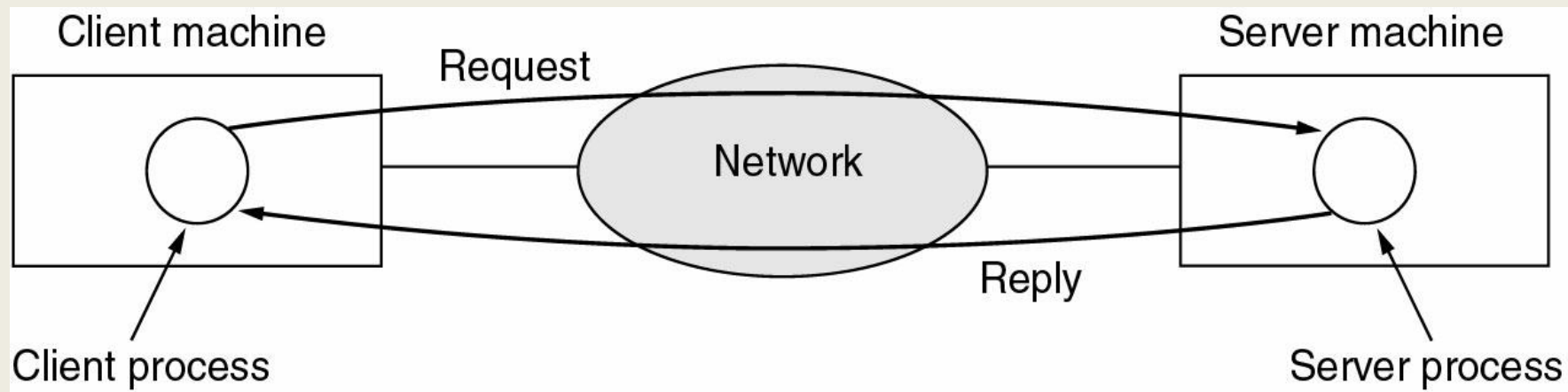
- Browsing websites using Web browsers.
- Accessing news, social media, educational content, and entertainment.
- Reading e-books, online journals, and digital libraries (ACM, IEEE, etc.)

This access operates primarily through **two models**:



Client–Server Model

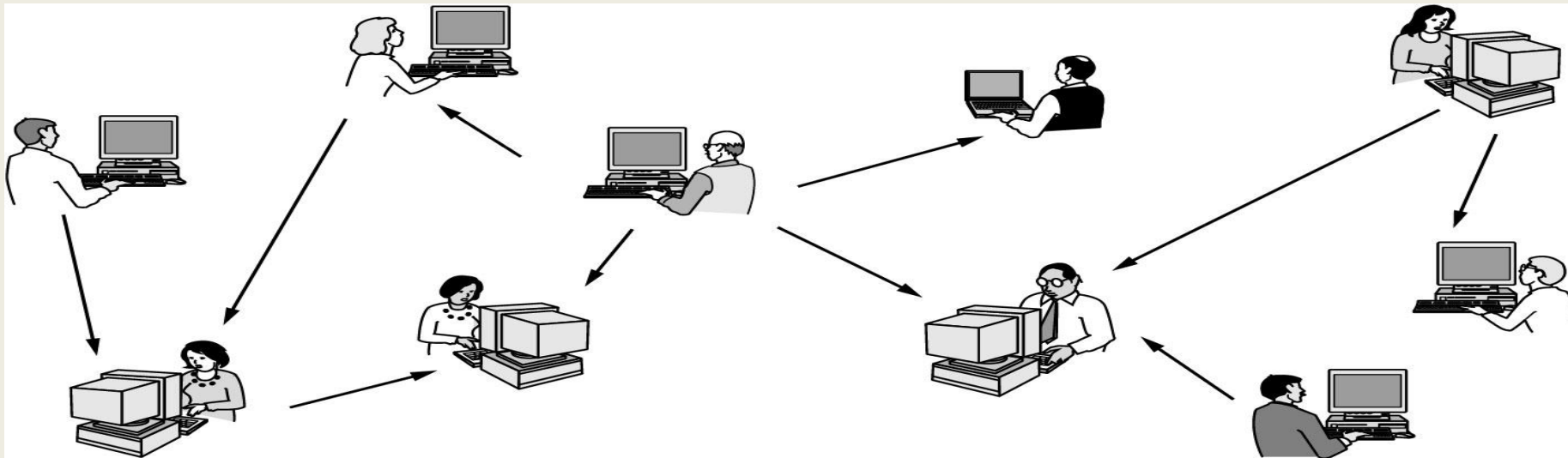
A client requests information while a server provides it. The server processes the request, fetches or generates the required data, and returns a response. This model supports scalable, controlled access to large volumes of information.



The client-server model involves requests and replies.

Peer-to-Peer Model

In this model, all devices act as equals, sharing information directly without a centralized authority. Each system contributes to and retrieves from the collective pool of data. This allows distributed and decentralized information access.



In peer-to-peer system there are no fixed clients and servers.

2. Person-to-Person Communication

- Computer networks enable direct communication between individuals, often in real time. This communication may involve text, audio, video, or multimedia.
- Networks support various forms of interpersonal communication, ranging from asynchronous methods (messages that can be read later) to synchronous methods (live interactions).

Forms of communication:

- **Email** -sends text, images, audio, video.
- **Instant Messaging/Chat** -WhatsApp, Messenger, UNIX talk.
- **Voice/Video Calls** -Skype, Zoom, VoIP.
- **Social Networking** -Facebook, Twitter, Instagram.

3. Electronic Commerce (E-Commerce)

E-commerce refers to conducting commercial and financial activities using computer networks. Networks allow transactions to occur electronically without physical presence.

It supports:

- **Business transactions**, where organizations and customers exchange goods, services, or information through digital means.
- **Financial operations**, such as payments, fund transfers, and investment activities through secure online systems.

Computer networks ensure secure communication, authentication, and data integrity, which are essential for completing economic activities online.

Types of e-commerce:

B2C (Business-to-Consumer): Amazon, Flipkart (ordering products online).

B2B (Business-to-Business): Companies ordering supplies from vendors.

G2C (Government-to-Consumer): Tax filing portals, bill payments.

C2C (Consumer-to-Consumer): OLX, eBay (selling used products).

P2P: Direct user-to-user sharing (like money transfer).

4. Entertainment

Networks have become a major platform for distributing digital entertainment. Instead of relying on physical media or traditional broadcast systems, content can be delivered on demand over networks.

This includes:

- Continuous delivery of multimedia content.
- Interactive services where users can control or influence the flow of content.
- Shared virtual environments where multiple users participate simultaneously.

Computer networks make entertainment accessible anywhere, anytime, and on multiple devices, enhancing user experience.

Examples:

- Streaming **music** (Spotify), **movies** and **videos** (Netflix, YouTube).
- Downloading media — songs, movies.

5. Internet of Things (IoT)

The Internet of Things refers to the integration of everyday objects into a network, allowing them to sense, collect, and exchange data. These devices operate with minimal human intervention.

Examples:

- Smart home devices: lights, thermostats, fridges, security cameras.
- Smart meters: electricity, gas, water usage reporting.
- Wearables: fitness trackers, smartwatches.
- Sensors in cars, industries, agriculture.
- Automated alerts (e.g., smoke detectors contacting fire department).

Types of Computer Networks

Computer networks exist in many forms depending on how they provide connectivity, how they are used.

1. Broadband Access Networks
2. Mobile and Wireless Access Networks
3. Content Provider Networks
4. Transit Networks
5. Enterprise Networks

1. Broadband Access Networks

- Broadband access networks provide high-speed Internet connectivity to homes and small offices.
- The primary purpose is to deliver fast and continuous connection so users can access information, communicate, and use online services.
- Broadband can be delivered through **copper cables**, **coaxial cables**, or **optical fiber**.
- The increasing demand for internet access has pushed broadband technologies to offer very high speeds, reaching gigabit levels in many regions.
- As more people gained home Internet connections, the usefulness and popularity of the Internet grew dramatically.

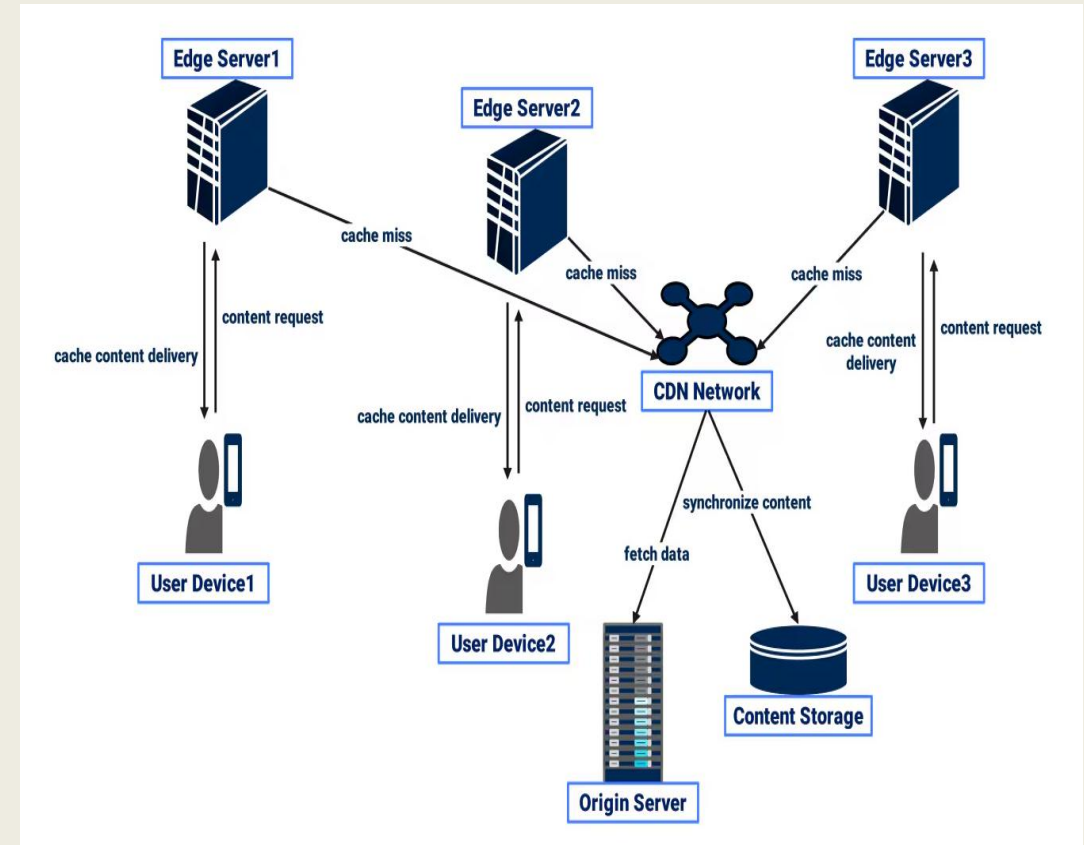
2. Mobile and Wireless Access Networks

- Mobile and wireless networks provide connectivity without requiring physical wired connections.
- They enable devices such as smartphones, tablets, and laptops to access the Internet from virtually anywhere.
- Wireless networks include:
 - **Cellular networks**, operated by telecom companies.
 - **Wi-Fi networks**, found in indoor or public spaces.
- Wireless networks support mobility, allowing users to access services while moving.
- They have become essential for modern communication, navigation, and portable computing.
- Wireless networking is also vital for industries, transportation, logistics, and scientific research, where devices or sensors must operate without fixed infrastructure.

3. Content Provider Networks (Data-Center Networks & CDNs)

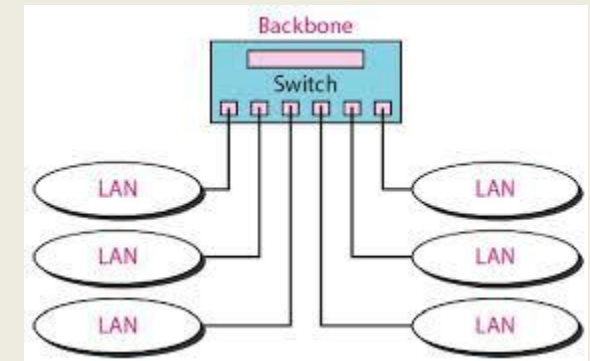
- A **data-center network** is a specialized network inside a massive facility that houses thousands or millions of servers.
- Its main job is to move large volumes of data efficiently between servers and to deliver that data to users on the Internet.
- Important challenges in data-center networks include:
 - **High throughput**, ensuring data can flow between any two servers quickly.
 - **Scalability**, as the number of machines increases.
 - **Fault tolerance**, so that the system continues to work even when some components fail.

- To ensure fast global access, many companies use **Content Delivery Networks (CDNs)**.
- A CDN distributes copies of frequently accessed content across many geographically spread servers.
- When a user requests data, the CDN serves it from the closest or least congested location, reducing delay and improving performance.



4. Transit Networks (Backbone Networks)

- Transit networks act as the intermediate carriers that transport data between different networks.
- When a user on one network accesses content stored in another, the data often travels through one or more transit networks.
- These networks form the **backbone of the Internet**, linking:
 - Access networks (used by consumers),
 - Data centers and content providers,
 - Other regional or national networks.
- However, as large Internet service providers and major content providers have grown, many now interconnect directly, reducing dependency on transit networks.
- Today, transit networks are mainly used as backup paths or when direct connections are not feasible.



5. Enterprise Networks

- Enterprise networks support the internal operations of an organization such as a business, university, or government office.
- They connect employees, departments, and various resources into a single unified communication system.

Key purposes of enterprise networks include:

a) Resource Sharing

Employees can access shared devices (printers, storage systems) and shared data (databases, documents) irrespective of their physical location.

b) Communication

Enterprise networks enable internal communication using email, VoIP (Voice over IP), video conferencing, and real-time collaboration tools. This allows teams across different locations to work together efficiently.

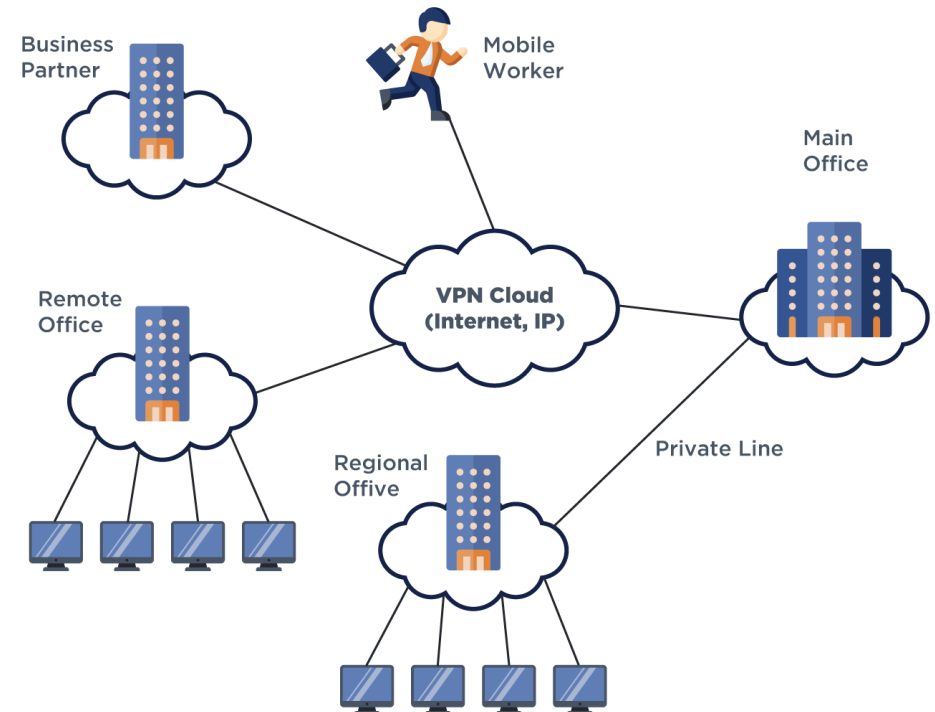
c) Remote Access via VPNs

Using Virtual Private Networks (VPNs), remote employees can securely access the company's internal resources as if they were physically present in the office. This removes geographical barriers and supports distributed work environments.

d) Electronic Business Operations

Organizations use enterprise networks to interact electronically with suppliers, customers, and partners. This includes inventory management, online ordering, and automated transactions, improving efficiency and reducing operational delays.

Enterprise WAN Network



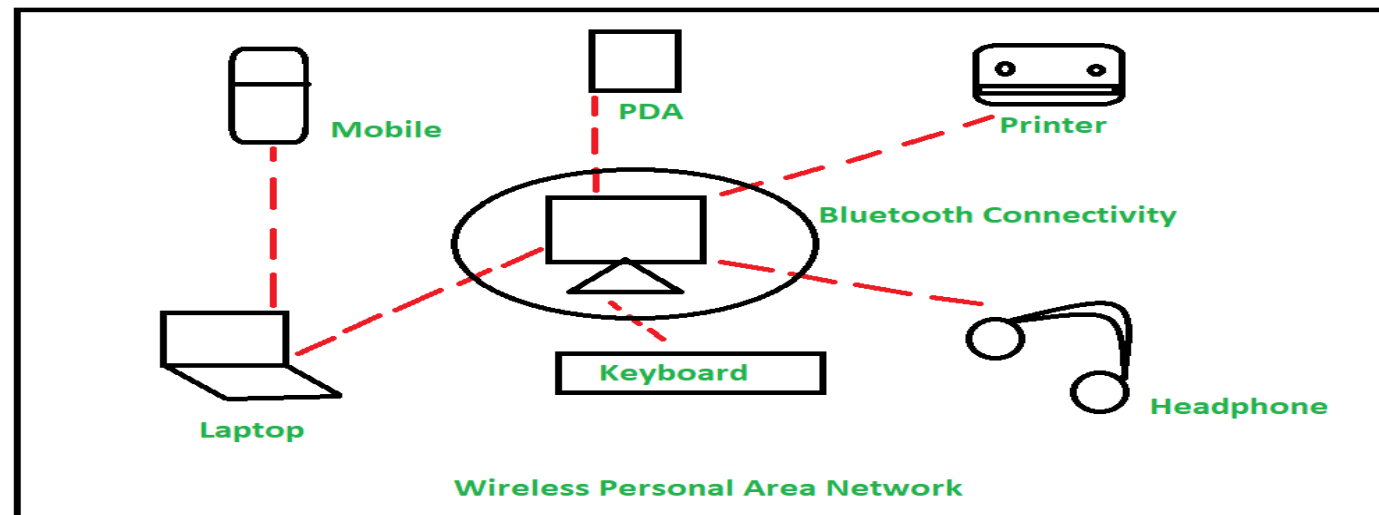
NETWORK TECHNOLOGY -FROM LOCAL TO GLOBAL

Each type of NETWORK TECHNOLOGY is defined by the geographic reach it serves, the devices it connects, and the design goals (latency, throughput, manageability, mobility, reliability, and security).

- Personal Area Networks (PANs)
- Local Area Networks (LANs)
- Metropolitan Area Networks (MANs)
- Wide Area Networks (WANs)
- Home Networks
- Transit / Backbone Networks and SD-WANs
- Internetworks

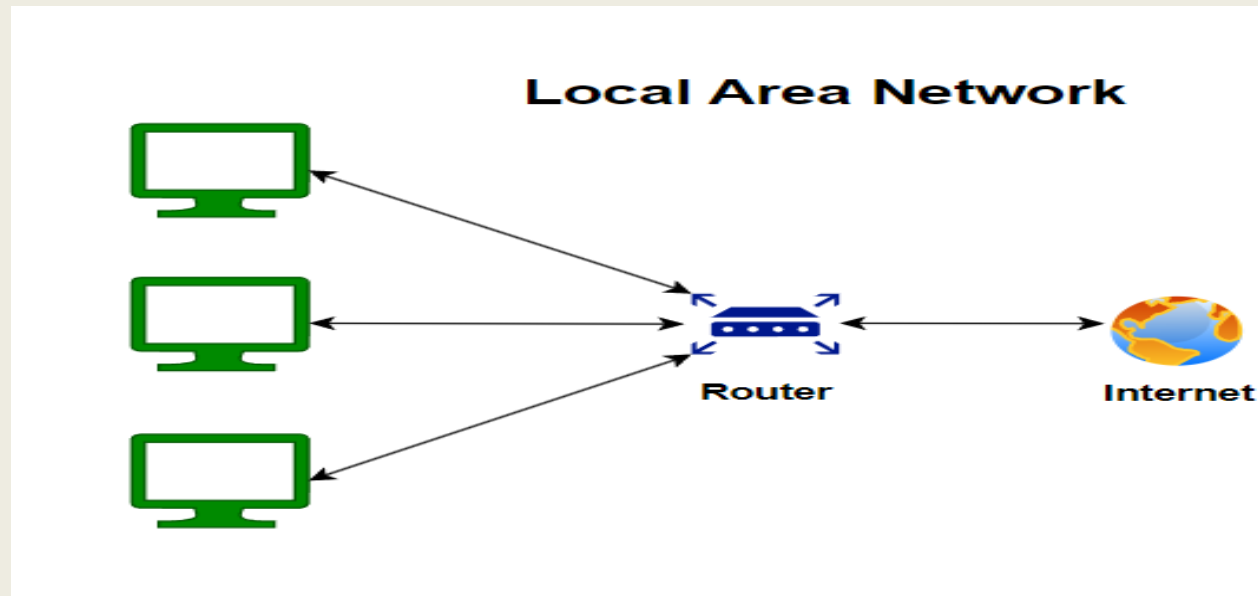
Personal Area Networks (PANs)

- PANs cover the shortest range -roughly the area around one person.
- Their purpose is to let personal devices (phones, headsets, wearables, keyboards, sensors) communicate seamlessly without wires.
- PAN technology emphasizes ease of use, very low power, and short-range operations.
- Control is commonly centralized (a master device coordinates timing and addressing) so devices can join and leave quickly and conserve energy.



Local Area Networks (LANs)

- LANs operate within a confined area such as a room, building, or campus.
- Their core goal is low-latency, high-throughput, and reliable sharing of resources (files, printers, applications) among many closely located hosts.
- LANs may be wired (switched point-to-point topologies) or wireless (access-point or mesh architectures).

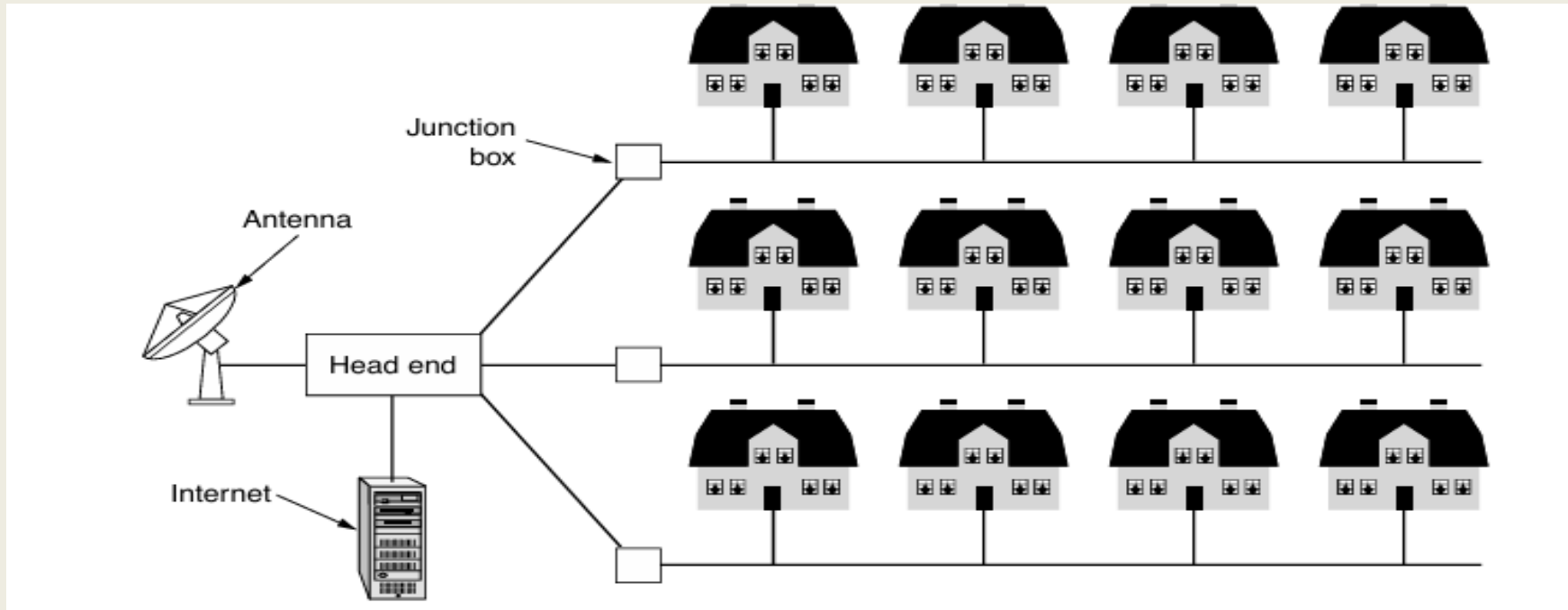


Home Networks

- A home network is a specific kind of LAN optimized for nontechnical users and a very broad variety of devices.
- It must balance **usability**, **interoperability** (heterogeneous devices working together), **security** (protecting privacy and safety), and **cost**.
- Because devices may remain connected for long periods, and because consumer devices prioritize low price, design choices often trade robustness and strong security for simplicity and low cost.

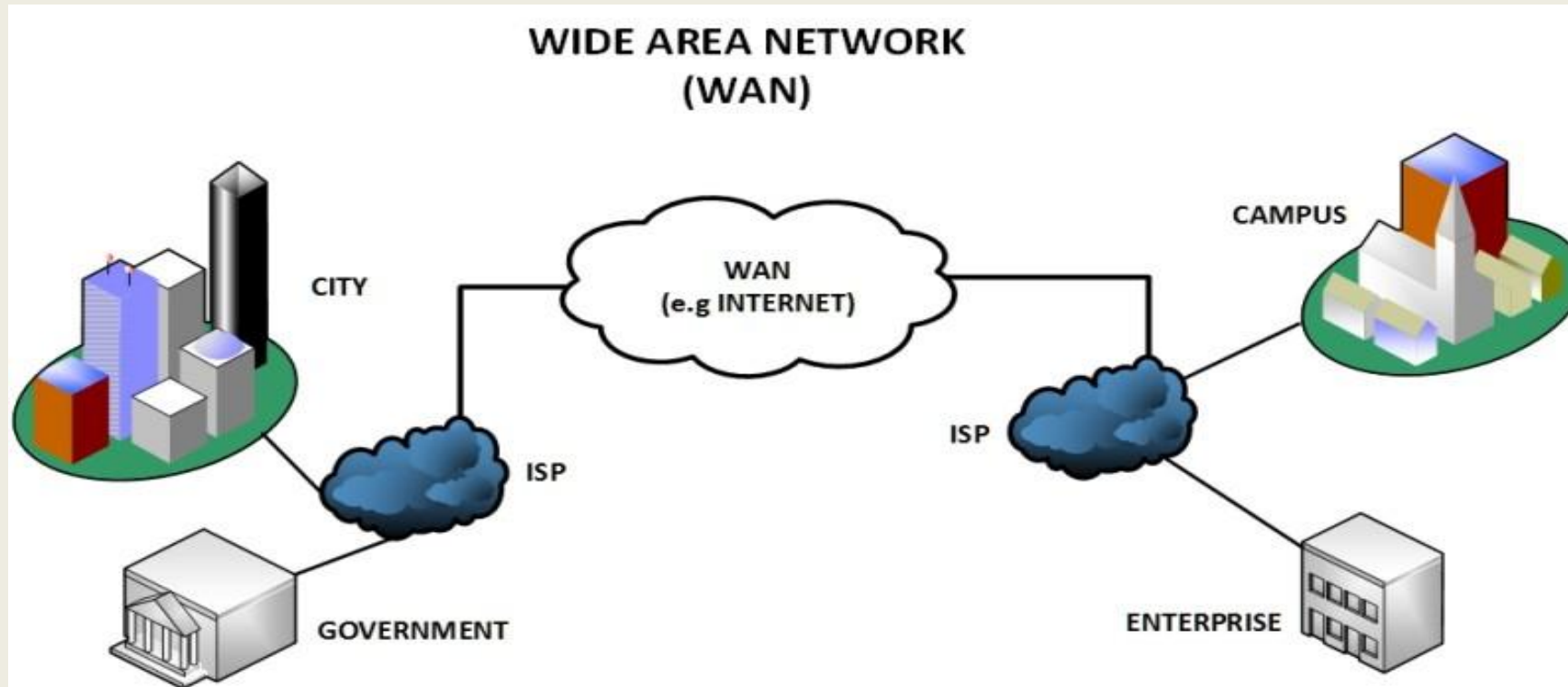
Metropolitan Area Networks (MANs)

- MAN, technologies must provide higher-capacity links and coordination across many subscribers while often reusing existing infrastructure (e.g., cable networks or purpose-built metropolitan fiber rings).
- They are intermediate in scale between LANs and WANs and are designed to aggregate traffic from many local access networks for distribution across the city.



Wide Area Networks (WANs)

- WANs connect geographically distant sites - cities, countries, or continents.
- They are composed of transmission links (media) and switching elements (routers).
- WAN design emphasizes long-distance capacity, heterogeneous link technologies, and routing across multiple possible paths.
- Unlike LANs, The WANs commonly involve different administrative domains, so issues of interconnection, service-level agreements, and heterogeneous technology translation are central.



- Virtualization techniques allow organizations to create logical private networks over shared infrastructure.

Virtualization is a technology that allows a single physical system (such as a computer, server, or network) to create and run multiple virtual versions of resources.

Transit / Backbone Networks and SD-WANs

- Transit (backbone) networks carry traffic between access networks and content providers.
- Historically essential for end-to-end connectivity, their role has changed as large content and access providers interconnect directly.
- Newer WAN design patterns use software control (SD-WAN / SDN) to combine multiple underlying paths and technologies and to programmatically optimize cost, performance, and resilience.

Internetworks

- An internetwork is a system formed by connecting two or more independent networks so, hosts on one network can communicate with hosts on another.
- The purpose is simple: let devices in different networks exchange data as if they were part of a single larger network.
 - **Composition:** An internetwork = many individual networks (LANs, WANs, MANs, etc.) joined together.
 - **Device that connects them:** A router (also called a gateway) forwards packets from one network to another and performs necessary protocol translation at the network layer.

- **Addressing & routing:** Each host has an address (e.g., an IP address).
Routers use routing tables and algorithms to find a path from source to destination across the internetwork.
- **Administrative domains:** Networks making up an internetwork are usually run by different organizations (ISPs, companies, universities). Each domain has its own policies (security, routing, billing).
- **Heterogeneity problem:** Networks can use different link technologies, speeds, and policies. The internetwork must hide those differences so end-to-end communication works.

Broadcast Networks

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	Local area network
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Classification of interconnected processors by scale.

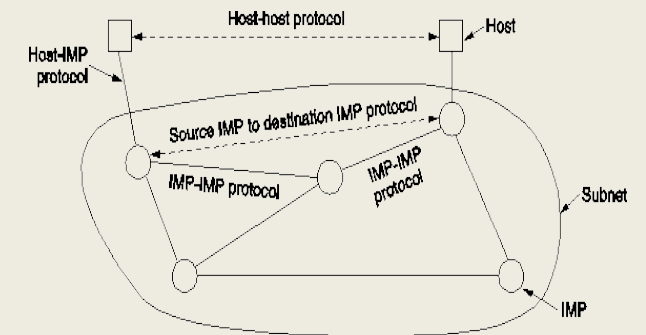
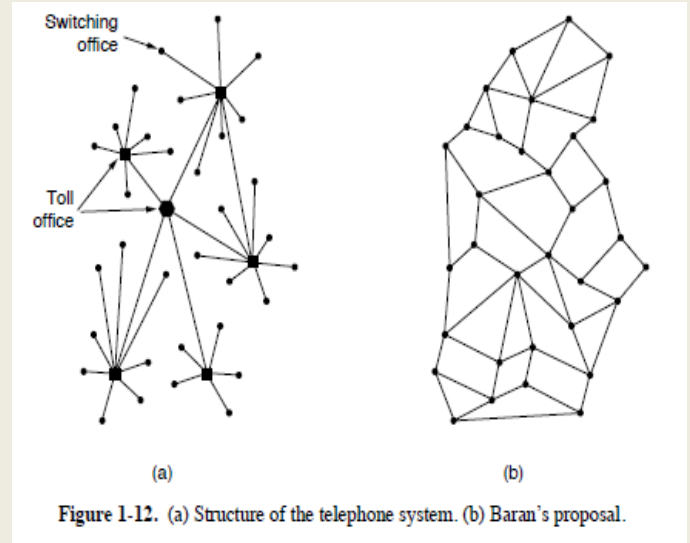
EXAMPLES OF NETWORKS

1. The Internet

The Internet a collection of independently operated networks that use common protocols (especially TCP/IP) so hosts on different networks can communicate.

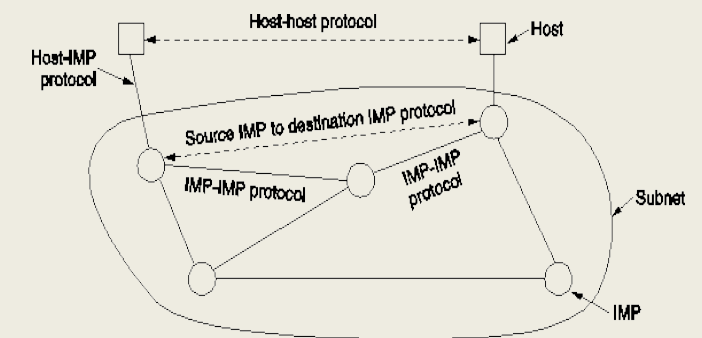
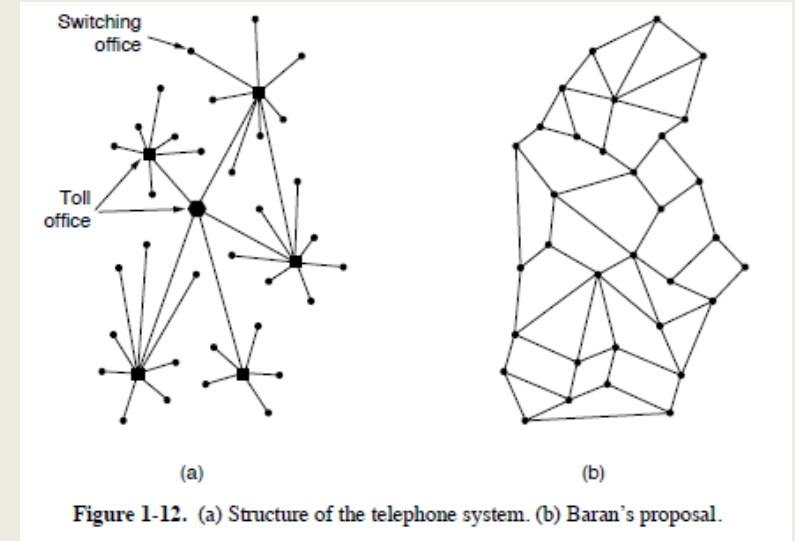
ARPANET:

- Early research led to **ARPANET** (late 1960s): a packet-switched network that introduced routers (IMPs) and store-and-forward packet delivery.
- Host could send messages of up to 8063 bits to its IMP, which would then break these up into packets of at most 1008 bits and forward them independently toward the destination.
- Each packet was received in its entirety before being forwarded, so the subnet was the first electronic store-and-- forward packet-switching network.

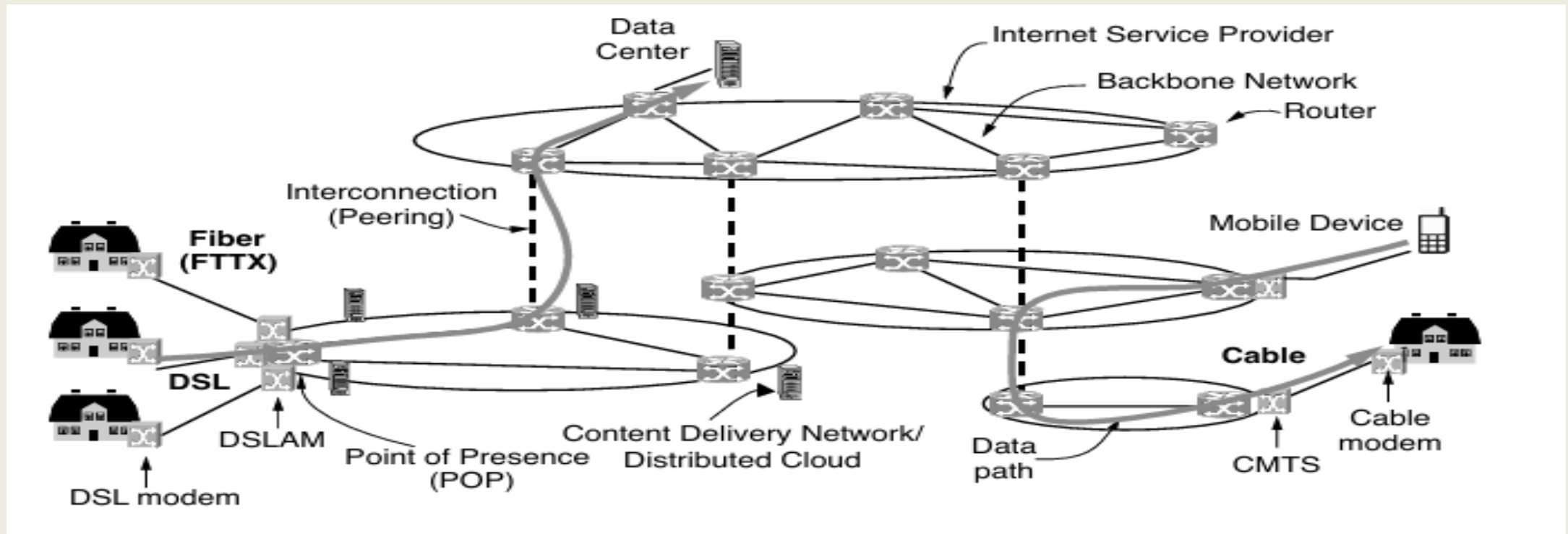


NSFNET:

- NSF decided to build a backbone network to connect its six supercomputer centers, in San Diego, Boulder, Champaign, Pittsburgh, Ithaca, and Princeton.
- Each supercomputer was given a little brother, consisting of an LSI-11 microcomputer called a fuzzball.
- The fuzzballs were connected with 56-kbps leased lines and formed the subnet, the same hardware technology the ARPANET used.
- NSFNET later expanded research networks and helped move the system toward a public, commercial Internet.



- Commercialization in the 1990s and the World Wide Web led to explosive growth; the Internet evolved from email and research tools to a global platform for web, media streaming, social networks, and cloud services.



Overview of the Internet architecture.

Architecture (high level):

- **Edge:** user devices (homes, mobiles) connect through an ISP.
- **Access networks:** last-mile technologies (DSL, cable, fiber, mobile).
- **Backbone / transit:** high-capacity networks that carry traffic between ISPs and data centers.
- **IXPs (Internet Exchange Points):** physical locations where ISPs peer and exchange traffic.
- **Content & data centers / CDNs:** host and replicate content close to users for lower latency and high throughput.
- Modern Internet topology: historically hierarchical (tier-1 → regional → access), but has **flattened** as large content providers and ISPs interconnect directly.

2. Mobile Networks

Networks that provide wireless connectivity over large areas using cellular base stations, optimized for mobility and broad coverage.

Architecture:

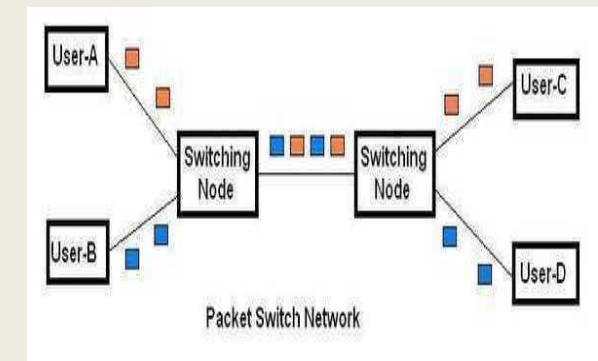
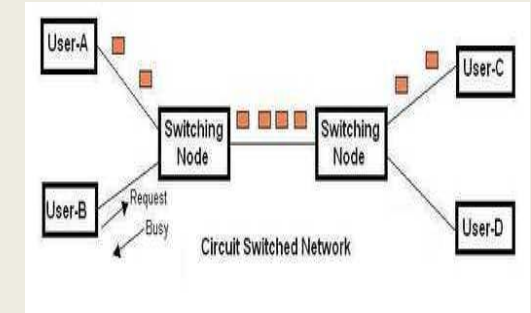
- **Radio access network (RAN):** base stations (cells) and protocols for over-the-air communication.
- **Core network:** handles switching, routing, subscriber management, and connectivity to external networks (Internet, PSTN).
- Important core elements (4G/LTE): eNodeB (base station), S-GW, P-GW; 5G adds new, flexible core functions.

Packet vs circuit switching:

- Traditional telephony used **circuit switching** (reserved path, simpler QoS).
- Internet uses **packet switching** (each packet routed independently, more resilient).

Generations:

- **1G:** analog voice, Low capacity, insecure.
- **2G:** digital voice, SMS, Better spectral efficiency.
- **3G:** packet data + voice, tens to hundreds of kbps.
- **4G (LTE):** packet-switched all the way; Mbps speeds, Improved latency.
- **5G:** higher throughput, lower latency, new frequency bands, supports massive IoT and ultra-low latency apps.



3. Wireless LANs (Wi-Fi)

- Provide **local wireless connectivity** within homes, offices, campuses, cafés, etc.
- Allow devices to access and share **LAN resources**, printers, and the **Internet** without cables.
- It provides flexibility, mobility within small areas, easy installation.



Security: Early WEP(Wired Equivalent Privacy)was weak and replaced by **WPA(Wifi Protected Access)/ WPA2 / WPA3** and stronger authentication frameworks.

Basic model:

- **Access point (AP)** mode: clients connect via APs which bridge to the wired LAN.
- **Ad-hoc** mode: direct client-to-client (less common).

Uses:

- Ideal for local-area wireless connectivity in homes, offices, campuses, cafés, etc.
- Provides high-speed access to LAN resources (printers, shared files, servers).
- Connects devices to the Internet without cabling.

Limitations:

- Not suitable for **wide-area mobility** (unlike cellular networks).
- Works only within **tens of meters**; walls and obstacles weaken signals.
- Weak or outdated setups are vulnerable to attacks.

NETWORK PROTOCOLS

- **Network protocols** are standardized sets of rules that enable communication between devices in a network.
- They define how data is formatted, transmitted, and processed, ensuring seamless interaction between devices, regardless of their hardware or software differences.

Design Goals

Networks aim to satisfy four main goals:

1. Reliability
2. Resource allocation
3. Evolvability
4. Security

1. Reliability

- Make the network work correctly even when parts fail
- Networks are made of many unreliable component's, wires get noisy, bits flip due to interference, routers crash, packets get lost.

To keep communication, correct, networks use:

- **Error detection** - Checksums / CRC detect if a packet is damaged.
- **Error correction**- Extra redundant bits let the receiver fix the error without retransmission.
- **Retransmission (ARQ)**- If a packet arrives damaged, the sender resends it.
- **Routing around failures**- If one route fails, packets automatically take another route.

2. Resource Allocation

- Fair and efficient sharing of network resources
- A network has limited resources like:
 - ❑ Bandwidth
 - ❑ Router buffer space
 - ❑ CPU time

Key issues:

- **Statistical multiplexing**- Share bandwidth dynamically based on demand (not fixed portions).
- **Flow control**- Prevent a fast sender from overwhelming a slow receiver.
- **Congestion control**- Prevent too many senders from flooding the network at once.
- **Quality of Service (QoS)**-Real-time apps (video calls) need timely delivery.

3. Evolvability

- Network must support future growth and new technologies
- Networks constantly grow as more devices join, new link technologies appear, and new routing methods are introduced.
- To support this, protocols must evolve without breaking existing systems.

Mechanisms include:

- **Layering**-Layering hides implementation details, allowing changes inside a layer without affecting the other layers.
- **Addressing/Naming**- Every device must be uniquely identifiable.
- **Internetworking**-Different networks have different packet sizes, ordering rules.

4. Security

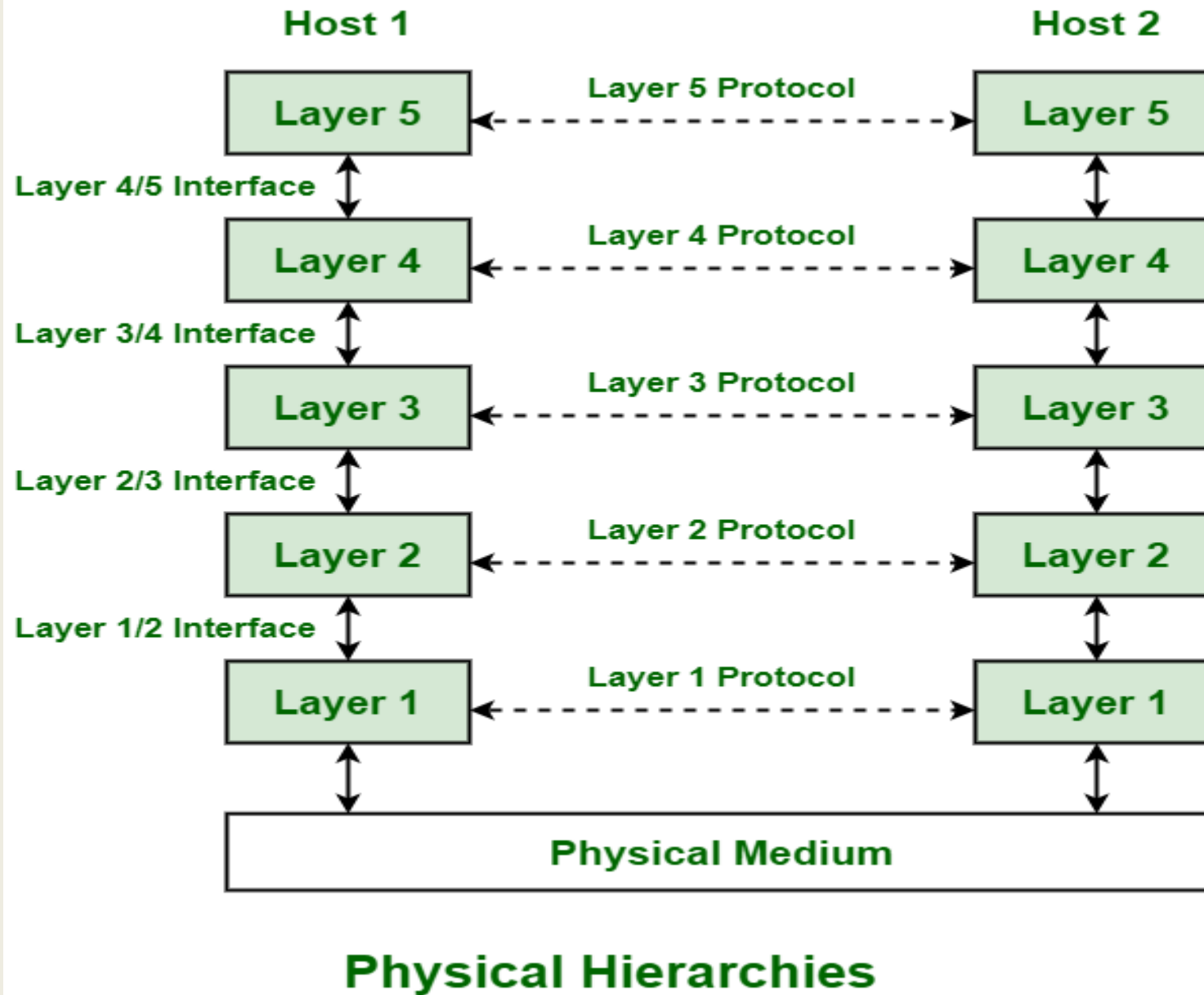
- Protect communication from attacks
- Threats include eavesdropping, impersonation, and message modification.

To protect users:

- **Confidentiality**-Encryption prevents outsiders from reading your data.
- **Authentication**-Ensures the sender/receiver is who they claim to be. e.g., SIM card in mobile networks
- **Integrity**-Ensures the message wasn't changed in transit.

Protocol Layering:

- Protocol layering is used in networks to reduce design complexity by organizing the system into multiple layers, with each layer built on top of the one below it.
- Every layer provides specific services to the layer above while hiding the internal implementation details, similar to concepts in computer science such as information hiding, data encapsulation, and abstract data types.
- When layer n on one machine communicates with layer n on another machine, they follow a common set of rules called the layer- n protocol, which defines how the communication should take place.
- The components at the same layer on different devices are called peers, and they use these protocols to interact.



Connections and Reliability

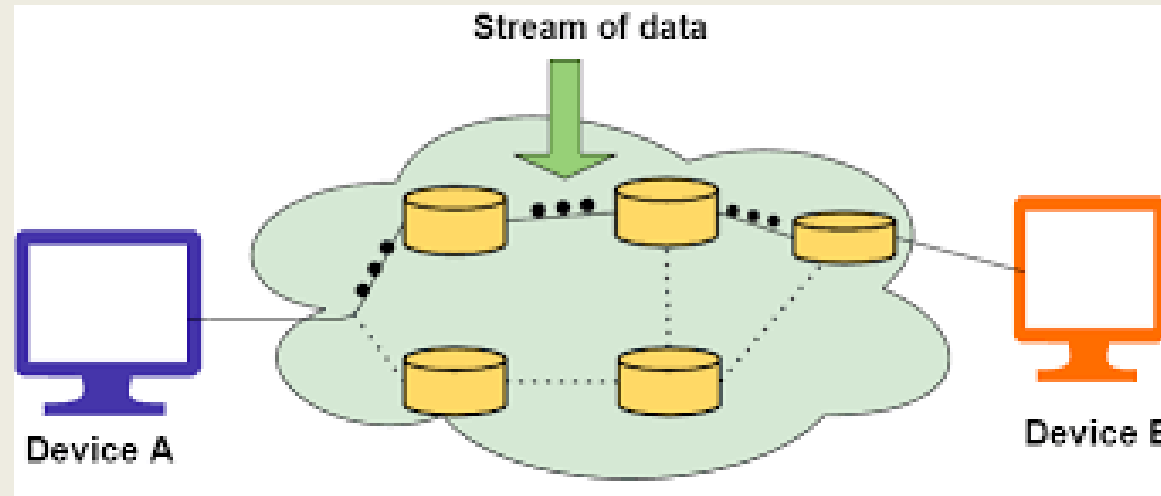
- Layers offer two types of service to the layers with various levels of reliability.
 - **connection-oriented**
 - **connectionless**

1. Connection-Oriented Service

- Connection-oriented service works like a telephone call.
- Before sending data, the sender and receiver first establish a connection.
- The connection behaves like a tube: data sent by the sender arrives at the receiver in the same order.
- During connection setup, both sides may negotiate parameters such as message size, quality of service, or bandwidth.

Reliability in Service

- Reliability is usually **Connection-Oriented** high because the receiver sends **acknowledgements** for received data.
- If packets are lost or damaged, they are retransmitted.
- Useful for applications where accuracy and order are critical, such as **file transfer**.



Connectionless Service

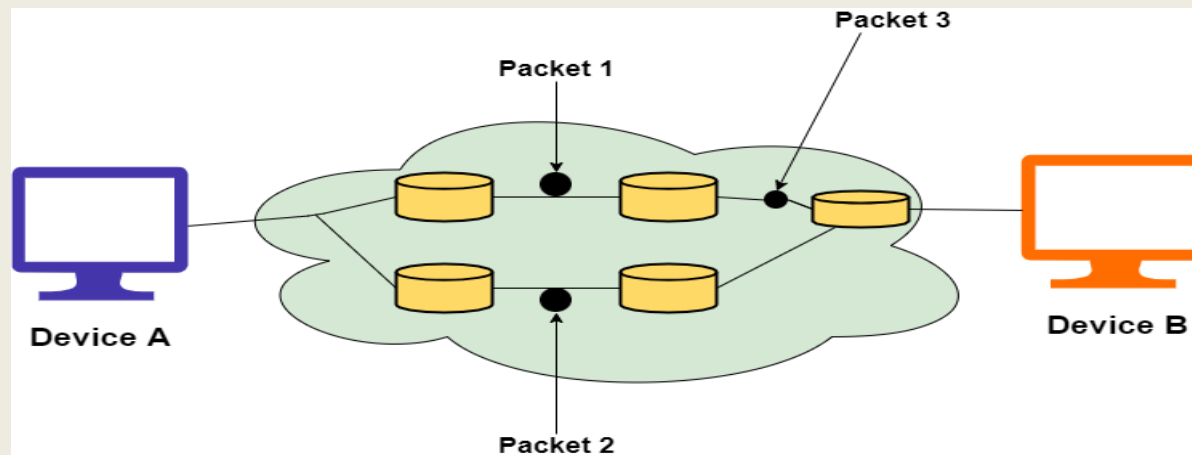
- **Connectionless service works like the postal system.**
- Each message (packet) carries its own destination address and is routed independently.
- No connection is set up before sending data.

Packets may arrive:

- Out of order
- Delayed
- Or even lost
- Techniques like **store-and-forward** or **cut-through switching** may be used inside the network.

Reliability in Connectionless Service:

- In connectionless service, reliability is optional.
- The basic form is an unreliable datagram service, where packets are sent without acknowledgements, so they may be lost, duplicated, or arrive out of order.
- Many real-time applications such as video conferencing prefer unreliable connectionless service because small errors are acceptable, while the delays caused by acknowledgements would interrupt the real-time experience.



Service Primitives

- Service primitives are the fundamental operations provided by a network layer that allow user processes to interact with the communication service.
- These primitives are usually implemented as system calls when the protocol stack is inside the operating system.
- Different services—connection-oriented and connectionless—provide and use different sets of service primitives because each service requires different operations for communication.
- For a reliable connection-oriented service, primitives like listen, connect, accept, receive, send, and disconnect are used to establish a connection, exchange data, and terminate it.

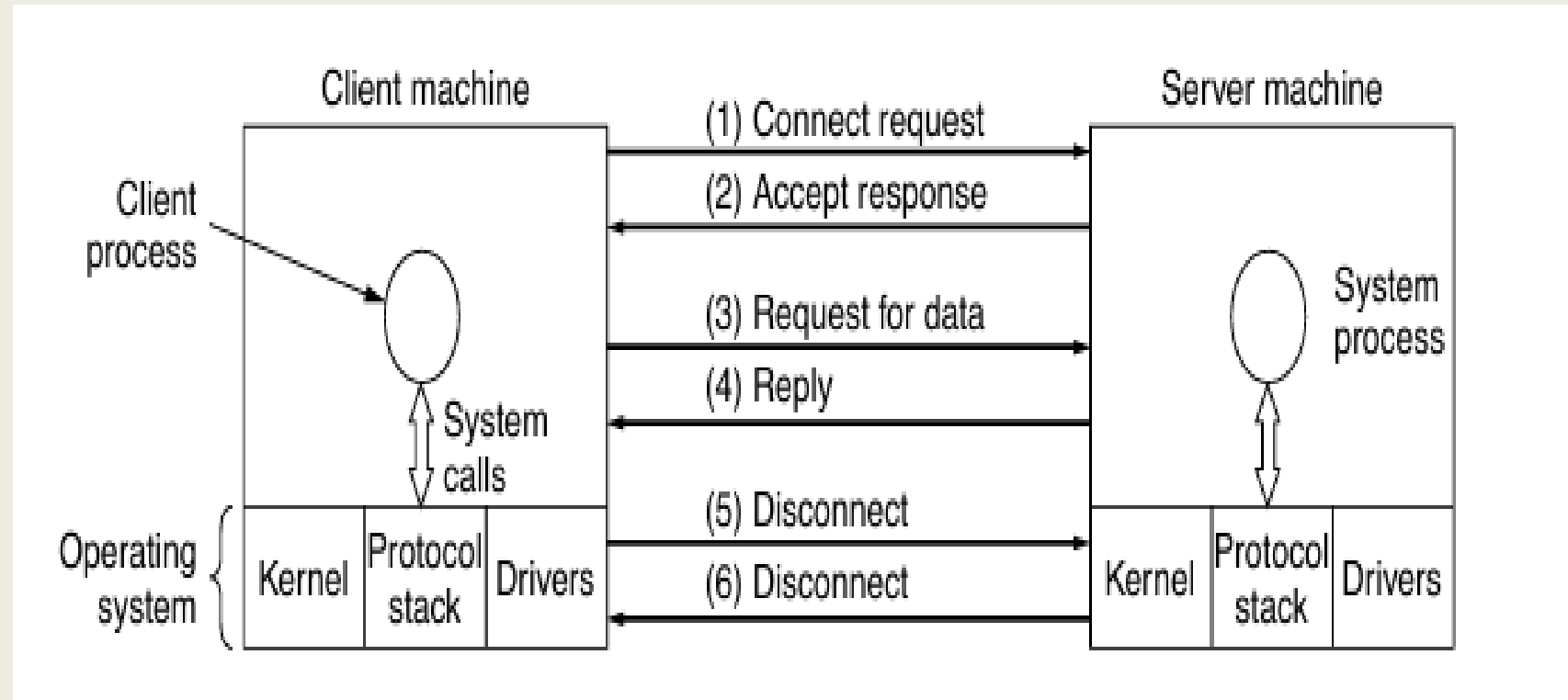
Six service primitives that provide a simple connection-oriented service

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Primitives Work in a Client–Server Model

- Consider a client–server interaction using **acknowledged datagrams**.
- The server begins by executing the ***LISTEN*** primitive, which tells the operating system that it is ready to accept incoming connections.
- The client begins communication by executing ***CONNECT***, specifying the address of the server. The operating system then sends a packet to the server requesting a connection.
- When the packet reaches the server machine, the OS checks if any process is waiting in the LISTEN state.
- If so, it unblocks that process, and the server calls ***ACCEPT***, which formally establishes the connection and sends a response packet to the client.
- Now both sides have an active connection and can proceed with actual data exchange.

- After the connection is set, the server typically executes **RECEIVE**, preparing to accept data from the client.
- The client sends its request using the **SEND** primitive, which unblocks the servers RECEIVE.
- The server processes the request and sends back a reply using SEND, causing the client's RECEIVE to return with the response.
- When the client is finished, it sends a **DISCONNECT** command. This usually blocks the client until an acknowledgement arrives.
- The server receives the disconnect packet, issues its own DISCONNECT, and the connection is fully closed.
- At this point, both sides return to their normal state.



A simple client-server interaction using acknowledged datagrams.

The Relationship of Services to Protocols

- Services and protocols are two different concepts in networking, even though they work together.
- A **SERVICE** is a set of operations (primitives) that a layer provides to the layer above it. It defines what the layer can do, without specifying how those operations are implemented.
- This service forms the interface between layers, where the lower layer acts as the service provider and the upper layer is the service user.
- A **PROTOCOL**, on the other hand, defines the rules, formats, and meaning of the messages exchanged between peer entities at the same layer on different machines.
- The relationship is that **protocols implement services**, but the service itself stays independent of the protocol.
- Because of this separation, protocols can be changed or upgraded internally without affecting the service seen by upper layers.

Reference models

- Reference models provide a structured way to design and understand network protocols by dividing the complex task of networking into manageable layers.
- Each layer performs specific functions and interacts only with the layers directly above and below it.

Two major layered designs dominate networking:

- OSI reference model, which provides a more detailed, standardized framework.
- TCP/IP reference model, which forms the foundation of the modern Internet.
- Both models aim to define the responsibilities of each layer and the way layers communicate.

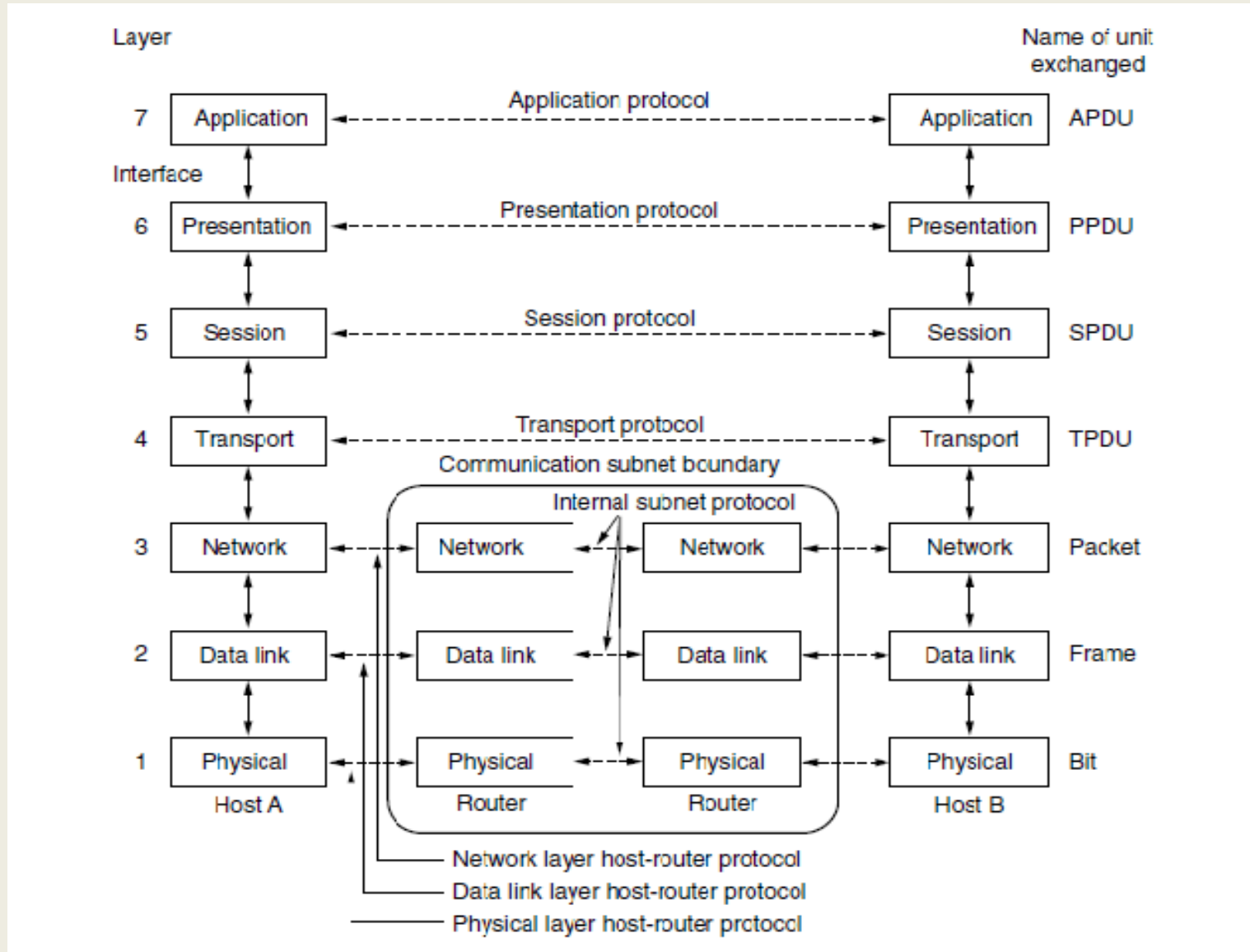
The OSI Reference Model

- The OSI (Open Systems Interconnection) model, developed by ISO, is a standardized framework that divides network communication into seven distinct layers.
- It was created to help different systems communicate using internationally agreed-upon protocols.
- The idea behind the seven layers is to separate networking into clear, logical functions, with each layer providing a specific service while hiding its internal workings from others.

The OSI model was designed by following simple rules:

- A layer should be created where a different abstraction is needed.
- Each layer should perform a well-defined function.
- The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
- The layer boundaries should be chosen to minimize the information flow across the interfaces.
- The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

The OSI Reference Model



PHYSICAL LAYER:

- The **physical layer** is concerned with transmitting raw bits over a communication channel.
- The **physical layer** focuses on ensuring that bits sent by one device are received correctly by another, meaning a transmitted **1** is received as **1**, not **0**.
- It deals with how bits are physically represented and transmitted, such as the type of electrical or optical signals used, how long each bit lasts, and whether data can flow in both directions at the same time.
- This layer also defines how connections are established and terminated, the number of pins in network connectors, and the function of each pin.
- Overall, the physical layer handles **mechanical, electrical, and timing details**, as well as the actual transmission medium used to carry data.

DATA LINK LAYER

- The **data link layer** is responsible for making a raw physical transmission channel appear reliable to the network layer.
- Its main task is to **detect and handle transmission errors** so that higher layers do not see corrupted data. To achieve this, the sender divides the data into **frames** and sends them one by one.
- If reliable service is used, the receiver sends an **acknowledgement** for each correctly received frame, allowing the sender to retransmit frames if errors occur.
- Another important function of the data link layer is **flow control**, which prevents a fast sender from overwhelming a slower receiver.
- It regulates the rate of transmission so the receiver can process data comfortably.
- This is handled by a special part of the data link layer called the **Medium Access Control (MAC) sublayer**, which manages access to the shared medium.

NETWORK LAYER

- The **network layer** is responsible for controlling how data moves through the network. Its primary function is **routing**, which means deciding the path that packets should take from the source to the destination.
- Routing decisions can be based on fixed routes that rarely change, routes that automatically adjust when parts of the network fail.
- Routes chosen at the start of a communication session, or highly dynamic routes that are recalculated for each packet based on current network conditions.
- Another important responsibility of the network layer is **congestion control**. When too many packets exist in the network at the same time, congestion can occur, leading to delays and bottlenecks.
- The network layer works together with higher layers to reduce traffic and manage network load. Additionally, the network layer plays a key role in providing **quality of service (QoS)**.

- The **transport layer** is responsible for providing reliable communication between end systems. Its main function is to accept data from the upper layers, divide it into smaller units if necessary, and pass them to the network layer for transmission.
- At the receiving end, it ensures that all data pieces arrive correctly and are reassembled in the proper order.
- The transport layer also decides the **type of service** to offer to applications, such as a reliable, error-free, point-to-point connection where data is delivered in the same order it was sent.
- Since the transport layer carries data directly from the source host to the destination host, it is considered a **true end-to-end layer** in the network architecture.

SESSION LAYER

- The **session layer** enables users on different machines to establish and manage communication sessions between them.
- It provides **dialog control**, which decides whose turn it is to transmit data during communication.
- The session layer also handles **token management**, ensuring that only one party performs a critical operation at a time, preventing conflicts.
- Additionally, it supports **synchronization** by inserting checkpoints in long data transfers so that, if a crash occurs, communication can resume from the last checkpoint instead of starting over.

PRESENTATION LAYER

- The **presentation layer** deals with the **format, structure, and meaning** of the data being transmitted.
- It ensures that computers with different internal data representations can communicate by defining data in an **abstract, standard format** before transmission.
- This layer handles **data encoding, decoding, and translation**, allowing complex data structures such as text, images, or banking records to be exchanged correctly between systems.

APPLICATION LAYER

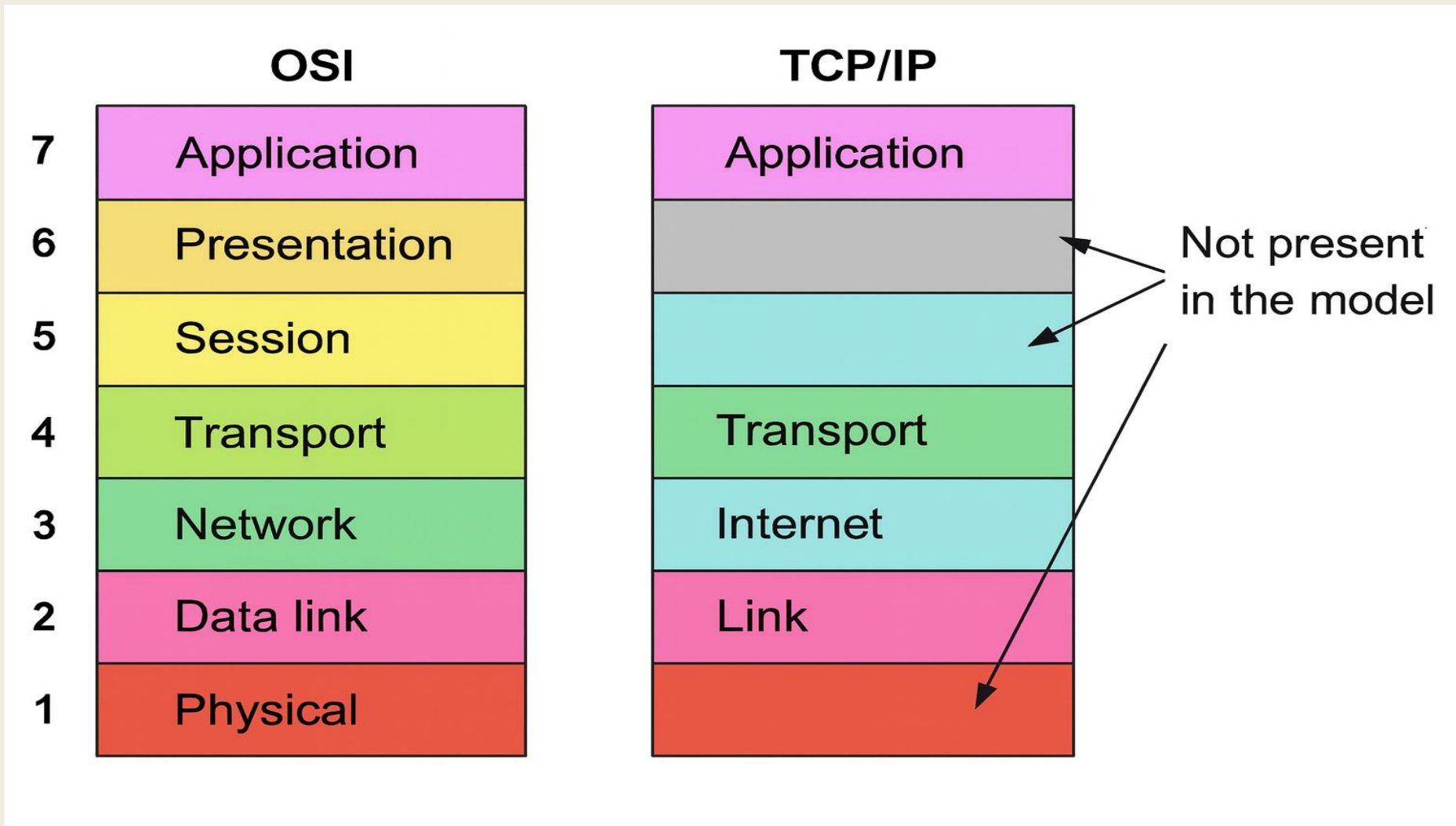
- The **application layer** provides network services that are directly used by end users and applications.
- It contains a variety of protocols that support common user activities such as accessing web pages, transferring files, sending emails, and remote login.
- A widely used application-layer protocol is **HTTP**, which forms the foundation of the **World Wide Web**, while other protocols enable file transfer, electronic mail, and remote system access.

Three concepts are central to the OSI model:

- Services
 - Interfaces
 - Protocols
- **Services-** A service is the function that a layer offers to the layer above it.
 - **Interfaces-** An interface defines how one layer uses the services of the layer below it.
 - **Protocols-** A protocol is the set of rules that defines how corresponding layers on different machines talk to each other.

TCP/IP Reference Model

- The TCP/IP reference model was created for the ARPANET, the early network funded by the U.S. Department of Defense (DoD), and later became the basis of the entire Internet.
- TCP/IP was specifically designed for **internetworking**, meaning it could connect many different networks and make them work as one.
- The key goal of TCP/IP was ensuring that communication should continue **even if parts of the network fail**, as long as the sender and receiver are still functioning.
- It supports different kind of applications with different requirements.
- This design made TCP/IP robust, flexible, and ideal for the global Internet we use today.



The Link Layer

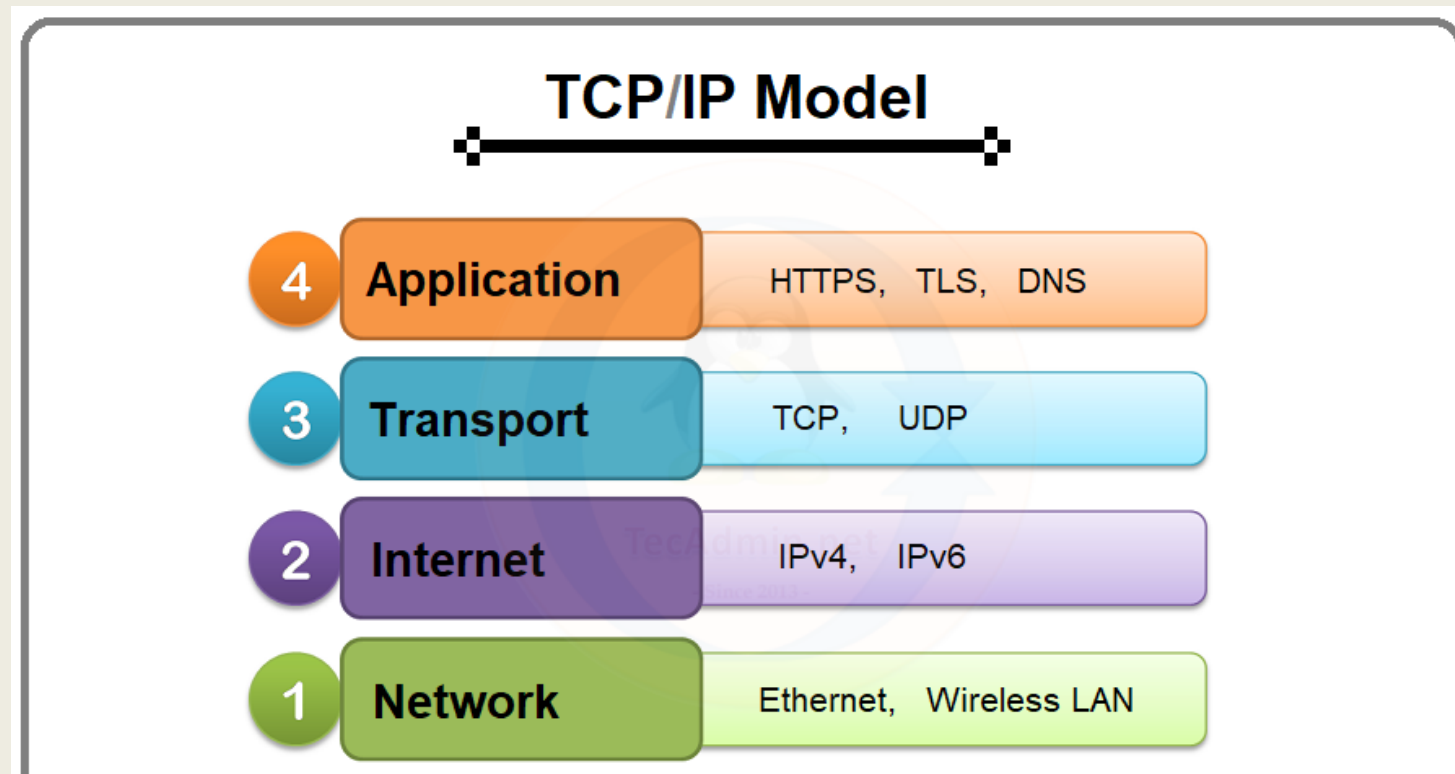
- The link layer is the lowest layer of the TCP/IP model.
- It was created to support the main goal of TCP/IP- **connectionless, packet-switched internetwork** that works across many different physical networks.
- Unlike the layers in the OSI model, the link layer is **not a fully defined protocol layer**.
- Instead, it acts as a **bridge** between the host (computer) and the physical transmission media (like Ethernet cables, Wi-Fi, fiber, or serial links).
- Its job is to handle whatever functions are needed so that packets can be sent to the next node on the network.

Internet Layer

- The **internet layer** is the central part of the TCP/IP architecture.
- Its main job is to allow hosts to send packets across **any type of network** and ensure that those packets can travel independently to the destination even if the destination is on a completely different network.
- Packets may take different paths and may even arrive in a different order than they were sent.
- It works like an **international postal system** each packet is routed independently, and the internal details of each network are hidden from users.
- This layer uses **IP(Internet Protocol)** for addressing and routing, and ICMP(Internet Control Message Protocol) for error reporting.

Transport Layer

- The **transport layer** in the TCP/IP model sits above the internet layer and enables communication between processes running on different computers.
- It provides end-to-end communication, similar to the transport layer in the OSI model.



TCP/IP defines two main transport protocols:

1. TCP (Transmission Control Protocol)

- Reliable and connection-oriented
- Ensures data is delivered accurately and in order
- Splits data into segments, sends them to the internet layer, and reassembles them at the destination

2. UDP (User Datagram Protocol)

- Unreliable and connectionless
- Does not guarantee order, delivery, or error-free transmission
- Used in cases where speed is more important than reliability, such as video streaming, online gaming, or voice calls

Application Layer

- The **application layer** is the topmost layer in the TCP/IP model.
- Unlike the OSI model, TCP/IP does **not** include separate session or presentation layers.
- Instead, application layer that self handle any session management or data formatting they require.

The application layer contains all **high-level protocols** that users interact with.

- **TELNET** – virtual terminal access
- **FTP** – file transfer
- **SMTP** – email

A Critique of the OSI Model and Protocols

Neither the **OSI model** nor the **TCP/IP model** is perfect, and both have been widely criticized.

Why OSI did not take over the world ?

- Bad timing
- Bad technology
- Bad implementations
- Bad politics

Bad timing

- A standard succeeds only if it is introduced at the right time.
- If it comes **too early**, the technology is not well understood; if it comes **too late**, companies have already invested in other solutions and ignore it.
- The OSI protocols were introduced **after TCP/IP was already widely in use**, especially in universities.
- Since vendors did not want to support two protocol stacks, OSI failed to gain acceptance and never became popular.

Bad Design

- Another reason OSI failed is **poor design**. The seven-layer structure was chosen more for **political reasons than technical ones**.
- Some layers, like the **session and presentation layers**, do very little, while others, such as the **data link and network layers**, try to do too much.
- The OSI model and its protocols are also **very complex**, difficult to understand, and hard to implement efficiently.
- Many functions like **addressing, flow control, and error control** are repeated across multiple layers, which makes the system inefficient.
- Because of this complexity and inefficiency, OSI was less practical compared to simpler models like TCP/IP.

Bad Implementations

- Because the OSI model and its protocols were **very complex**, the first implementations were **large, slow, and difficult to use**.
- People who tried them had bad experiences, so OSI quickly gained a reputation for **poor quality**.
- Even though later versions improved, the negative image remained, and users were no longer interested.
- In contrast, early **TCP/IP implementations**, especially the one included with **Berkeley UNIX**, were **simple, efficient, and free**.
- This encouraged many users to adopt TCP/IP, which led to a growing user community and continuous improvements.
- As a result, TCP/IP gained popularity through positive feedback, while OSI declined due to its poor initial implementations.

Bad Politics

- Another reason OSI failed was **bad politics**. In the 1980s, **TCP/IP** was **closely associated with UNIX**, which was very popular and trusted in academic and research communities.
- OSI, on the other hand, was seen as a standard **pushed by government bodies**, especially European telecom authorities and later the U.S. government.
- Many developers felt that **bureaucrats were forcing a complex and inferior standard** on people who were actually building networks.
- This negative perception reduced support for OSI, similar to earlier cases where governments tried to impose languages like **PL/I or Ada**, which also failed to gain widespread acceptance.

A Critique of the TCP/IP Reference Model and Protocols

- Although TCP/IP became widely successful, it also has several weaknesses.
- First, the TCP/IP model does **not clearly separate services, interfaces, and protocols**, which is considered poor software design practice.
- Unlike the OSI model, TCP/IP does not clearly distinguish between what a layer offers and how it is implemented, making it less useful as a guide for designing new network architectures.
- Second, the TCP/IP model is **not general-purpose**. It works well only for TCP/IP–based networks and cannot easily describe other technologies such as Bluetooth.

- Third, the **link layer is not a true layer** in TCP/IP; it functions more as an interface, and confusing layers with interfaces weakens the clarity of the model.
- Fourth, TCP/IP does not separate the **physical layer** and **data link layer**, even though they perform very different functions.
- A proper reference model should treat them as separate layers. Finally, while core protocols like **IP and TCP** were well designed, many early supporting protocols were created in an ad hoc manner.
- Some of these outdated protocols, such as **TELNET**, are still in use today despite being poorly suited for modern computing environments.